MD. MEHERAB HOSSEN

Mechanical Engineer | CFD & Thermal Sciences

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Research Profile

Mechanical Engineering graduate (AUST, 2026) with experience in multiphase CFD and heat transfer.
Develop and validate ANSYS Fluent and OpenFOAM models against published benchmarks.
Research includes phase change, thermal energy storage, and VOF-based droplet dynamics.
Seek to extend this work to modeling flow and heat transfer in porous media and subsurface systems.

Education

Ahsanullah University of Science and Technology (AUST) **April 2022 – July 2026**
B.Sc. in Mechanical Engineering CGPA: 3.122/4.00

Manuscripts & Conferences

Journal of Energy Storage **First author**
"Numerical Investigation of Enhanced Thermal Energy Storage in Solid-Solid Phase Change Materials with New Fin Architecture for Shell-and-Tube Heat Exchanger Applications."
Submitted October 2026; under review.

10th BSME International Conference on Thermo-Fluid Engineering
Abstracts accepted; full papers submitted September 2026.

1. **Authored (first author):** Extending Weber–Ohnesorge Regime Maps to Oblique and Shear-Driven Droplet Impact on UAV Surfaces: A VOF-Based Parametric Study.
2. **Coauthored (second author):** Coupling Pre-Impact Thermal History to Resolved Freezing Dynamics of Free-Falling Supercooled Droplets on UAV-Relevant Cold Surfaces.
3. **Coauthored (third author):** Thermal Energy Storage Enhancement Using Alphabet-Inspired Fins in a Shell-and-Tube Heat Exchanger with Solid–Liquid PCM.

Research Experience

Undergraduate Research Assistant – Thermal Energy Storage **2025 – 2026**
Advisor: Prof. Dr. Dewan Hasan Ahmed, AUST

- Designed bio-inspired straight-fin arrangements for a shell-and-tube solid–solid PCM storage unit; modeled charging and discharging in ANSYS Fluent using temperature-dependent apparent heat capacity with a transition at 25 °C.
- Validated [validation quantity] against Farulla et al. and Alnakeeb et al., with a maximum deviation of 1%.
- Confirmed grid and time-step sensitivity within 1.57% for [sensitivity quantity].
- Compared five fin angles and orientations with an unfinned baseline; the best simulated design reduced charging time by 79.3%.

Research Assistant – Droplet Dynamics **2026 – Present**
Research mentor: Nowroze Farhan Ifraj, PhD student in Mechanical Engineering, Virginia Tech

- Investigate Weber-number effects on droplet spread ratio using the VOF method.
- Reproduce Rafi et al. (2022) falling-droplet deformation and collision cases in ANSYS Fluent, with a spread-ratio error of 3%. [Validation]
- Benchmark OpenFOAM droplet-pair impact simulations against Goswami and Hardalupas (2023) over We = 54–128, with an 8% deviation from published data. [Validation] [Videos]

Professional Experience

CFD Simulation Engineer **July 2026 – Present**

MACPROTEC, Houston, Texas, USA | Remote

- Analyze industrial dust collectors, baghouses and process pipelines using ANSYS Fluent, ANSYS Rocky and OpenFOAM.
- Develop 3D geometry in ANSYS DesignModeler and SolidWorks for simulation workflows.
- Reconstruct CAD geometry from laser scans using FARO SCENE, CloudCompare, MeshLab and Geomagic Design X.
- Configure discrete phase model (DPM) simulations of air–coal and air–dust particle transport.

Thesis

Experimental Characterization of *Nymphaea rubra* (Water Lily) Natural Fiber and Its Epoxy Composite through Mechanical Analysis **July 2026**

Supervisor: Dr. Fazle Rabbi, AUST | Three-member research team

- Investigated water-lily fiber as a sustainable reinforcement for epoxy composites; conducted ASTM-based single-fiber tensile testing using a universal testing machine.
- Identified 48-hour retting as the best extraction condition. Composite properties: tensile strength 25.5 ± 2 MPa; elastic modulus 3.2 ± 0.29 GPa; flexural strength 22 ± 5.5 MPa. [Research] [Lab gallery]

Projects

- Analyzed the **NACA 23012 airfoil** at $\mathrm{Re} = 2 \times 10^6$ over angles of attack from -5° to 24°; compared results with Miley's (1982) wind-tunnel data. *Tools: XFLR5, Microsoft Excel.*
- Estimated 76,733 kWh/year and a 3.26-year payback for a 50 kW grid-connected PV plant. [Report]

Skills

CFD & simulation: ANSYS Fluent, OpenFOAM, ANSYS Rocky, COMSOL.
CAD & reverse engineering: SolidWorks, ANSYS DesignModeler, FARO SCENE, CloudCompare, Geomagic Design X, MeshLab.
Programming: C, C++, MATLAB, Python.
Scientific communication: LaTeX, Microsoft Word, Excel (advanced), PowerPoint.

Awards & Training

- Completed BITAC training in CNC/lathe machining, heat treatment, molding, and casting. [Certificate]
- Earned the **People's Choice Award – Mechanical Project Showcasing, Fall 2023.** [Portfolio]
- Completed **MATLAB Onramp – MathWorks.** [Certificate]

References

Prof. Dr. Dewan Hasan Ahmed
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